Order what will sell, not what sold last month

Sales forecasting and reorder recommendations for online shops

Immanuel Albrecht · `https://github.com/alb-i/demand-showcase`

The problem

Every week you decide how much of each product to reorder. A rule of thumb such as "the last four weeks' sales plus a safety margin" is easy, but it reacts late: it runs out when demand picks up before a season and keeps too much when demand falls. Both cost money, as missed sales or as cash tied up in stock.

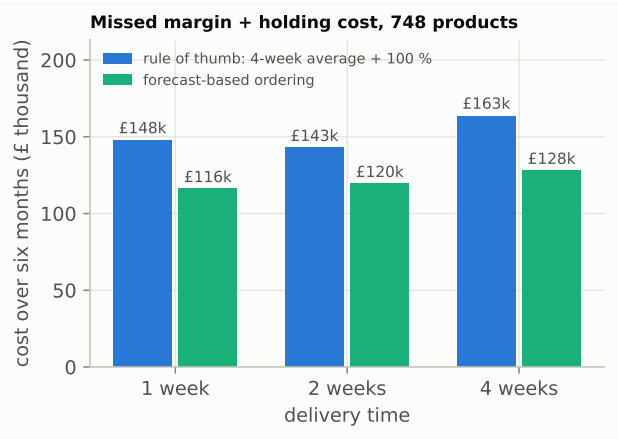
The approach in three steps

1. **Clean the sales history.** Returns, cancelled orders, one-off bulk orders and closed weeks are separated out, so that the weekly demand per product is what customers actually wanted.
2. **Forecast a range, not a number.** For every product: how much is likely to sell over the next weeks, and how much more or less is still plausible. The range matters more than the single best guess, because stock is there to cover the weeks that turn out higher.
3. **Turn the range into an order.** You choose a service target; each week the order tops the stock up to the amount that covers the demand until the next delivery with that probability. Before anything changes, a simulation replays your past months with your current rule and with the new one, so you see the difference in pounds.

A result on a real shop

A UK online giftware wholesaler (public data, 2009–2011[1]), its 748 best-selling products, over a six-month test period that was not used during development. With weekly orders and one week delivery time, the rule "four-week average plus 100 %" cost **£147.7k** in missed margin and holding cost and served 90.5 % of the demand. Forecast-based ordering cost **£116.3k** and served 93.0 %: **£31k less in six months** (21 %), with more customers served. At the same service level it needs 15–33 % less stock.

Assumed: 40 % margin, holding cost 25 % of the stock value per year. The gain comes from knowing the uncertainty: as a single number, the forecast is no better than the four-week average. Very high service targets (above about 95 %) get expensive.



What you provide

- A sales export: order lines with date, product, quantity and price (ideally two years).
- Delivery times per supplier and how often you order.
- Margins or purchase prices; your current reorder rule, if you have one.
- Optionally current stock levels, to start the order list.

What you get

- A weekly forecast per product, with its range.
- A weekly order list (CSV) that your team or system can use directly.
- Your current rule against the new policy, simulated on your own history, in your currency.
- A dashboard to change the service target and see stock and cost move, and a short written report.

[1] Chen, D. (2012). *Online Retail II* [Dataset]. UCI Machine Learning Repository. `https://doi.org/10.24432/C5CG6D`. Licensed under CC BY 4.0, `https://creativecommons.org/licenses/by/4.0/`. Changes: cleaned and aggregated to weekly demand per product.